72%

36/50

ABERDARE TECHNICAL COLLEGE

INFORMATION COMMUNICATION TECHNOLOGY MODULE 3

UNIT: NETWORK DESIGN AND MANAGEMENT

UNIT CODE: 0612 451 07A
PRACTICAL ASSESSMENT – OBSERVATION CHECKLIST

Time 4 hours Practical 2

A. CANDIDATE AND ASSESSOR DETAILS

Particular	Details
Candidate Name	MARTIN NDUGIRE
Candidate Registration No.	520
Assessment Date	
Start Time	8:30 am
Assessor Name	James
Assessment Venue	ICT lab

B. ASSESSMENT INFORMATION

Assessment Decision

The candidate shall demonstrate the ability to plan, install, configure, test and troubleshoot a small wired computer network using the equipment provided.

The assessor shall observe the candidate performing the practical activities and award marks based on demonstrated competence.

CamScanner

C. MATERIALS, TOOLS AND EQUIPMENT / CUTTING LIST

No.	Item / Material / Equipment	Quantity	Provided By
1	Desktop computers	3	Centre
2	USB printer with USB cable	1	Centre
3	8-port or 24-port switch	1	Centre
4	Router / wireless router with internet access	1	Centre
5	Cat5e/Cat6 UTP straight-through patch cables	5	Centre
6	RJ45 connectors	As required	Centre
7	Crimping tool	1	Centre
8	Cable stripper	1	Centre
9	Cable tester	1	Centre
10	Diagram sheets	As required	Centre
11	Network labels	As required	Centre
12	Pen/pencil/ruler	As required	Candidate

D. PRACTICAL OBSERVATION CHECKLIST – 50 MARKS

Instructions to Assessor:
Observe the candidate while performing each task. Award marks only for work demonstrated by the candidate. Use the comments column to record evidence, errors, corrections or observations.

No.	Item to be evaluated	Allocated Marks	Candidate Score	Assessor Comments
TASK A – PHYSICAL NETWORK DIAGRAM				
1	Draws a neat physical network diagram showing HR-PC, ACC-PC, REC-PC, switch, router, USB printer and ISP/Internet cloud.	2	02	neat, physical diagram
2	Correctly connects the three computers to the	2		

CamScanner

	central switch and connects the switch to the router.		02	comps connected to central switch
3	Shows the USB connection between the printer and REC-PC and identifies REC-PC as the print host.	2	02	usb connection shown
TASK A SUBTOTAL		**6**	06	
TASK B – LOGICAL TOPOLOGY DESIGN				
4	Draws the logical topology showing IP addresses, subnet mask, default gateway and internet path.	2	02	- drawn logical topology - shows IP and subnet - show gateway
5	Correctly shows how HR-PC and ACC-PC access the shared printer through REC-PC.	1	01	Access to printer shown
6	Identifies the physical topology as star and the logical topology appropriately.	1	01	Physical topology identified as star
7	Gives valid reasons for selecting the topology, such as easy fault isolation, expansion and centralised management.	1	00	Reason not stated
TASK B SUBTOTAL		**5**	04	
TASK C – ADDRESSING PLAN				
8	Identifies the network as 192.168.10.0/24 and gives the correct subnet mask.	1	01	Network id identified
9	Assigns 192.168.10.1 as the router LAN address and uses a static address.	1	01	Router assigned the IP address
10	Assigns unique IP addresses to HR-PC, ACC-PC and REC-PC within the correct subnet.	3	03	unique IP addresses assigned to PCs

CamScanner

11	Assigns the correct default gateway to the computers.	2	02	gateway assigned to PC
12	Includes the switch management address where applicable and provides a documented growth/reserved address range.	1	00	Reserved Range not provided / indicated
TASK C SUBTOTAL		**8**	07	
TASK D – INSTALLATION AND CONFIGURATION				
13	Connects the network devices correctly using Cat5e/Cat6 straight-through cables and labels cables/ports.	2	02	
14	Correctly demonstrates preparation/testing of a straight-through cable where required and obtains a successful tester result.	1	01	
15	Configures the router LAN IP, hostname and management password correctly.	2	01	Password not set
16	Configures the required DHCP/NAT/internet-sharing settings where applicable.	1	00	not set
17	Renames the computers as HR-PC, ACC-PC and REC-PC and places them in the BRIGHTPATH workgroup.	3	03	Renamed Placed in the BrightPath workgroup
18	Correctly configures the IP address, subnet mask and default gateway on the computers.	1	01	Configured correctly
19	Connects the USB printer to REC-PC, installs the driver and enables printer sharing.	1	01	Connected driver installed

CamScanner

20	Adds the shared printer to HR-PC and ACC-PC and demonstrates successful printing.	1	0/1	added the shared printer
21	Creates and shares CompanyShare so that all three computers can access it.	1	0/1	Creates shared files in shared files
22	Creates FinanceHR and configures access so that HR and Accounts can access it while Reception is denied.	1	0/1	NOT configured.
TASK D SUBTOTAL		**14**	11	
TASK E – TESTING AND VERIFICATION				
23	Tests connectivity between HR-PC and ACC-PC using ping and obtains successful replies.	1	0/1	
24	Tests connectivity between ACC-PC and REC-PC using ping and obtains successful replies.	1	0/1	
25	Tests connectivity between REC-PC and HR-PC using ping and obtains successful replies.	1	0/1	
26	Tests connectivity from each computer to the default gateway.	1	0/1	
27	Verifies that the shared printer is visible from HR-PC and ACC-PC.	1	0/0	NOT verified.
28	Verifies internet/ISP connectivity from all three computers.	2	0/2	verified
29	Performs a tracert test and correctly identifies the first hop as the default gateway.	1	0/0	NOT used!

CamScanner

30	Prints a test page from the required computers and records the result.	1	0	Printed a test page
31	Tests access to CompanyShare from all three computers.	1	0	tested access
32	Demonstrates that Reception is denied access to FinanceHR and records the result.	1	00	not demonstrated
TASK E SUBTOTAL		**11**	08	
TASK F – TROUBLESHOOTING				
33	Identifies the network symptom and follows a systematic troubleshooting procedure.	2	0	Procedure not followed
34	Uses appropriate diagnostic tools/commands such as ipconfig, ping, tracert and physical inspection.	1	0	tools and commands used
35	Correctly identifies the cause of the introduced fault.	1	0	identified
36	Corrects the fault without causing unnecessary changes to the network.	1	00	not corrected
37	Re-tests the affected connection/service and confirms that the network is working.	1	0	Re-tests
TASK F SUBTOTAL		**6**	04	
TOTAL PRACTICAL OBSERVATION MARKS		**50**	36	

E. PRODUCT CHECKLIST – 10 MARKS

Instructions to Assessor:
Inspect the completed network installation and other products produced by the candidate. Award marks according to the quality and functionality of the final products.

CamScanner

No.	Product / Expected Outcome	Allocated Marks	Candidate Score	Assessor Comments
1	Neat and correctly labelled physical network diagram showing all required devices, connections, ports and cable types.	2	02	neat physical diagram
2	Complete and accurate addressing table using the 192.168.10.0/24 network, including gateway and reserved growth range.	2	02	Network id used to complete the addressing table
3	Functional wired star network with correct device connections and working network links.	1	01	functional network
4	Functional shared USB printer accessible to the required computers through REC-PC.	1	01	functional usb printer
5	Functional CompanyShare folder accessible by all three computers.	1	01	
6	Functional FinanceHR folder with access restricted to HR and Accounts and Reception denied.	1	00	Reception not denied
7	Completed network test log with accurate Pass/Fail results and supporting test evidence.	1	½	
8	Neat cable/port labelling and organised network installation.	1	00	Cable not labelled
TOTAL PRODUCT MARKS		10	7½	

CANDIDATE FEEDBACK

Candidate Feedback on the Assessment

.....

.....

Candidate's Strengths

CamScanner

- Network design and Implementation
- Troubleshooting VLAN also Good.

Candidate's Areas for Further Improvement

.....

.....

MARK SUMMARY

Assessment Component	Maximum Marks	Candidate Score
Practical Observation Checklist	50	36
Product Checklist	10	7/h
TOTAL	60	

Percentage: _____ %

L. COMPETENCE / DECISION OUTCOME

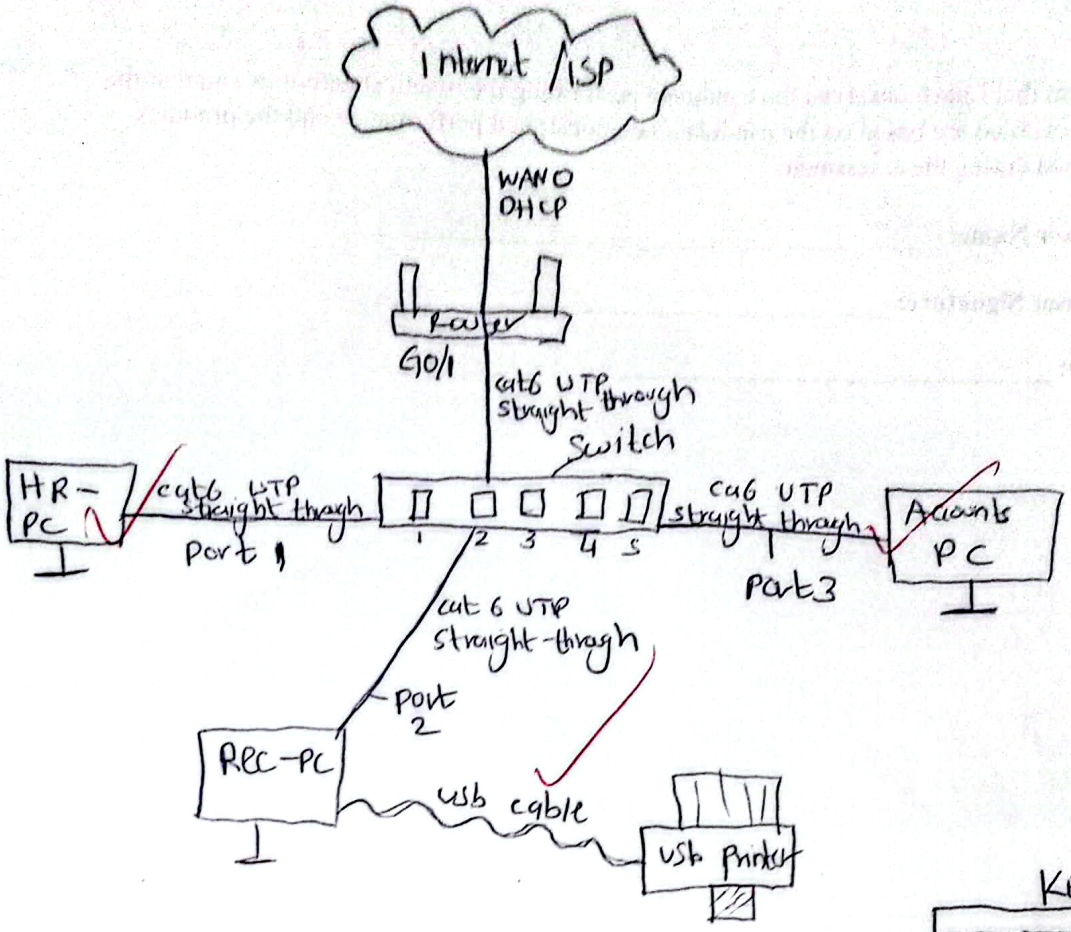
Decision	Tick
Competent	☑
Not Yet Competent	☐
Re-assessment Required	☐

Particular	Name / Signature	Date
Candidate Name		
Candidate Signature		
Assessor Name	James	
Assessor Signature		
Internal Verifier / Moderator		

ASSESSOR DECLARATION

CamScanner

Physical Network diagram



KEY

cat 6 UTP straight through

usb cable

Name : Martin

Date :

Logical topology design